Kaniuar Bacho

Kaniuar.Bacho@edu.rub.de Kaniuar Bacho Website

Education

M.Sc. Cyber Security, Ruhr University Bochum (RUB), *Germany* 2022 - 2024

- Grade: 96%/100% (excellent)
- Master's Thesis: "Compiling Nonlocal Games without QFHE"
- Advisors: Prof. Dr. Michael Walter & Prof. Dr. Giulio Malavolta
- Relevant modules:
  - Discrete Mathematics (*grade 100%*)
  - Information Theory (*grade 100%*)
  - Intro to Cryptography I + II (*both grade 100%*)
  - Cryptography (*grade 100%*)
  - Cryptographic Protocols (*grade 98%*)
  - Zero-Knowledge Proof Systems
  - Post-Quantum Cryptanalysis (*grade 100%*)
  - Quantum Cryptography (*grade 100%*)
  - Quantum Circuits (*grade 100%*)
  - Quantum Information and Computation
  - Quantum Algorithms Seminar
  - How to Build a Deep-Tech Start-up Seminar

B.Sc. Mathematics, University of Bonn, *Germany* 2017 - 2021

- Grade: 1.5/1.0 (very good)
- Special focus on Algebraic Number Theory and Algebraic Geometry
- Bachelor's Thesis: "Classification of Representations of $\mathrm{GL}_2(\mathbb{F}_{p^n})$" (*grade 1.1*)
- Advisors: Dr. Johannes Anschütz & Prof. Dr. Peter Scholze

Research and Work Experience

Quantum Technologies Intern | *TÜV Information Technology* Sep 2022

- Advised institutes on securing their Quantum Key Distribution setups against attack vectors
- Department-wide presentation on BB84 and contemporary side-channel attacks on QKD devices

Teaching Assistant | *Ruhr University Bochum & University of Bonn*

- Quantum Information and Computation (*Winter 2023/2024*)
- Cryptographic Protocols (*Summer 2023*)
- Analysis II (*Summer 2019*): The professor and my students graded my teaching with a 1.0

Mentor for Freshmen | *University of Bonn* Oct 2019 - Sep 2020

Visiting Researcher | *Max Planck Institute for Mathematics* Sep 2018

- Research on an open problem in Algebraic Topology in cooperation with other national winners of the German Federal Mathematics Competition

Honors & Awards

German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes)

- Germany's largest, oldest and most prestigious scholarship foundation
- Awarded full funding for my bachelor's and master's studies (>100,000€)

International Olympiad in Cryptography (NSUCRYPTO) Dec 2023

- Silver Medal, ranked 3rd
- Two of my solutions were selected to be the best solution

Federal Mathematics Competition (Bundeswettbewerb Mathematik) 2014 - 2017

- Two-time national winner 2016 & 2017 (my write-ups)
- Most prestigious mathematical competition in Germany, alongside the Mathematical Olympiad
- Awarded scholarship from the 'German Academic Scholarship Foundation'
- Interviewed on the German TV show 'WDR' and the radio

Mathematical Olympiad (German MO) 2010 - 2016

- Team member representing the German state of NRW for the finals in 2016
- Several first prizes in the first three rounds

Projects

Free Courses

- Quantum Complexity Theory by Sevag Gharibian (Course)
- Interactive Proofs with Quantum Devices by Thomas Vidick (Course)
- Quantum Key Distribution by Ramona Wolf (Course)
- Quantum Computation and Information by Ryan O'Donnell (Course)
- Cryptography by Dan Boneh (Course I & Course II)

Web Development (Personal website) 2023

- Developed my website using pure HTML and CSS

Hacking a Virtual Bank Apr 2023 - Jul 2023

- In the four-month university project, we explored hacking a virtual bank using techniques such as identifying logical flaws, XSS, SQL injections, local file inclusion, remote code execution, and CSRF

Machine Learning for Trading (Udacity course) 2023

- Course by Prof. Dr. Tucker Balch, Managing Director at J.P. Morgan AI Research
- Implementing machine learning-based strategies in Python to make trading decisions using real-world data

Talks

Slides of my talks are available at kaniuarbacho.github.io

Quantum Random Walks

- Quantum Algorithms Seminar at Ruhr University Bochum (*Jul 2023*)

Quantum Key Distribution

- Department-wide presentation at the company 'TÜV Information Technology' (*Oct 2022*)

Quadratic Forms over $\mathbb{Q}_p$

- Algebraic Number Theory Seminar at the University of Bonn (*Nov 2020*)

Classification of Representations of $\mathrm{GL}_2(\mathbb{F}_{p^n})$

- Defense of my Bachelor's Thesis at the University of Bonn (*Sep 2020*)

Representation Theory of Lie Algebras

- Lie Algebras Seminar at the University of Bonn (*May 2020*)

Random Walks on the Symmetric Group

- Markov Chains and Stochastic Algorithms Seminar at the University of Bonn (*Jul 2018*)

About Me

Programming	Python, SageMath, C/C++, HTML/CSS, Bash (Linux), Machine Learning for Trading
Languages	German (native), English (proficient, TOEFL Score 103)